OCR Error Recovery in Egyptian Medicine Search

A plain-English walkthrough of what works, what fails, and why

medicine-search-clean engineering analysis

14 July 2026

Bottom line

- **The number to quote: 46.34% Hit@1** and **70.91% Hit@20** on 464 scored, de-duplicated OCR-to-medicine pairs. This is the fair, one-vote-per-case score.
- **The number if you count every OCR output, repeats included:** 49.24% Hit@1 / 73.61% Hit@20 across all 595 rows.
- **It handles ordinary OCR typos well** (98% Hit@20 on standard errors) **and struggles hard on badly garbled text** (8% Hit@20 once more than 60% of the word is wrong).
- **The single biggest fixable problem:** predictions with more than one kind of character error mixed together (extra + missing + wrong letters in the same word) cause 84% of all misses.
- **Before tuning anything:** about 17–30 rows look like the OCR read a real, different medicine name correctly – these may be mislabeled ground truth, not search failures. Check them first (Section 12).

1 What was actually tested

- Photos of medicine packaging already went through an OCR model, producing a text guess (e.g. `MYMLAX`).
- That text guess – not the image – is fed into the search system ("Algorithm 4"), which searches a catalog of 25,066 medicines and returns a ranked list of likely matches.
- The correct answer (e.g. `MYOLAX`) is hidden from the search system and used only afterward, to score the result.
- **What this measures:** can the search step recover the right medicine from already-OCR'd text. **What it does not measure:** OCR accuracy itself, dosage reading, or anything about the original image.

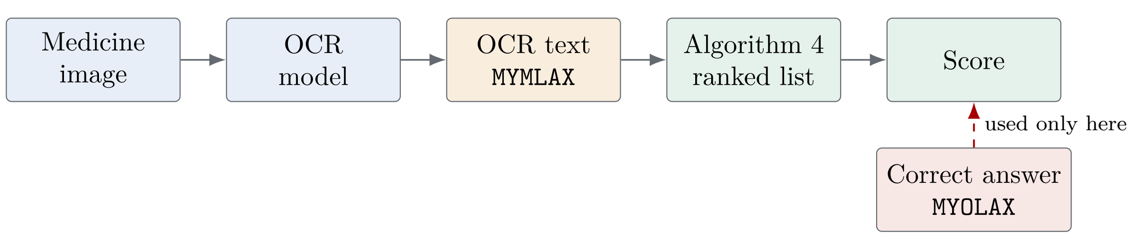

Figure 1: The correct answer never reaches the search step – it only checks the result afterward.

2 Which score to cite

There are four honest ways to count this dataset, and they give different numbers. Use the primary row for any claim about the algorithm; use the "all observations" row only when auditing raw data.

Denominator	Cases	Hit@1	Hit@20
All observations (every OCR output, incl. repeats)	595	49.24%	73.61%
Scored (17 real-drug-name conflicts removed)	578	50.69%	75.43%
Every unique query→target pair	477	45.07%	69.39%
Scored, unique pairs – PRIMARY	**464**	**46.34%**	**70.91%**

Table 1: Why the gap exists: 118 rows are the same medicine name typed by several different OCR models (one real case, several votes if you don't de-duplicate), and 17 rows are cases where the OCR text exactly names a *different* real medicine – a labeling conflict, not a search miss.

3 Terms used below, in plain English

- **Compact form**: lowercase the text and strip spaces/punctuation, so `CLEX A NE` and `CLEXANE` both become `clexane`.
- **Edit distance (d)**: number of single-character fixes (add/remove/swap one letter) needed to turn the OCR text into the target, after compacting. Example: `GHONT` → `CALAMINE` needs 7 fixes.
- **Normalized distance (r)**: d divided by the target's length. Example above: $7 \div 8 = 0.875$ – almost every letter is wrong.
- **Hit@k**: did the correct medicine appear anywhere in the top k results. Hit@1 = got it right on the first guess. Hit@20 = correct answer was somewhere in the top 20.
- **Risk tag on each row**: Safe (342 rows, 57%) – ordinary case; Caution (236 rows, 40%) – short query or $r > 0.40$, worth a second look; Dangerous (17 rows, 3%) – OCR text exactly names a different real medicine.

4 Dataset at a glance

Panel A – Split. Each medicine family lives entirely on one side, never both:

- **Development** – 475 rows (79.8%), used for analysis and tuning. Example: all 92 `KETOROLAC` rows are here.
- **Holdout** – 120 rows (20.2%), set aside to check the results hold on medicines never tuned on. Example: all 32 `VIGOREX` rows are here.

Panel B – Cohort (each row is checked against these rules top to bottom; the first one that matches wins):

- **Real-drug-name collision** – 17 rows (2.9%): the OCR text isn't garbled at all – it's the exact spelling of a *different* real medicine. Example: OCR reads `RIVOTRIL` but the labeled target was `RIVO`. This is a possible mislabel, not a search failure.
- **Normalized exact match** – 6 rows (1.0%): once spacing/punctuation is stripped, the OCR text matches the target exactly. Example: `TO CO` vs `TOCO`. Should always be found.
- **Extreme-distance prediction** – 121 rows (20.3%): more than 60% of the target's letters would need to change. Example: `GHONT` vs `CALAMINE`. The hardest group.

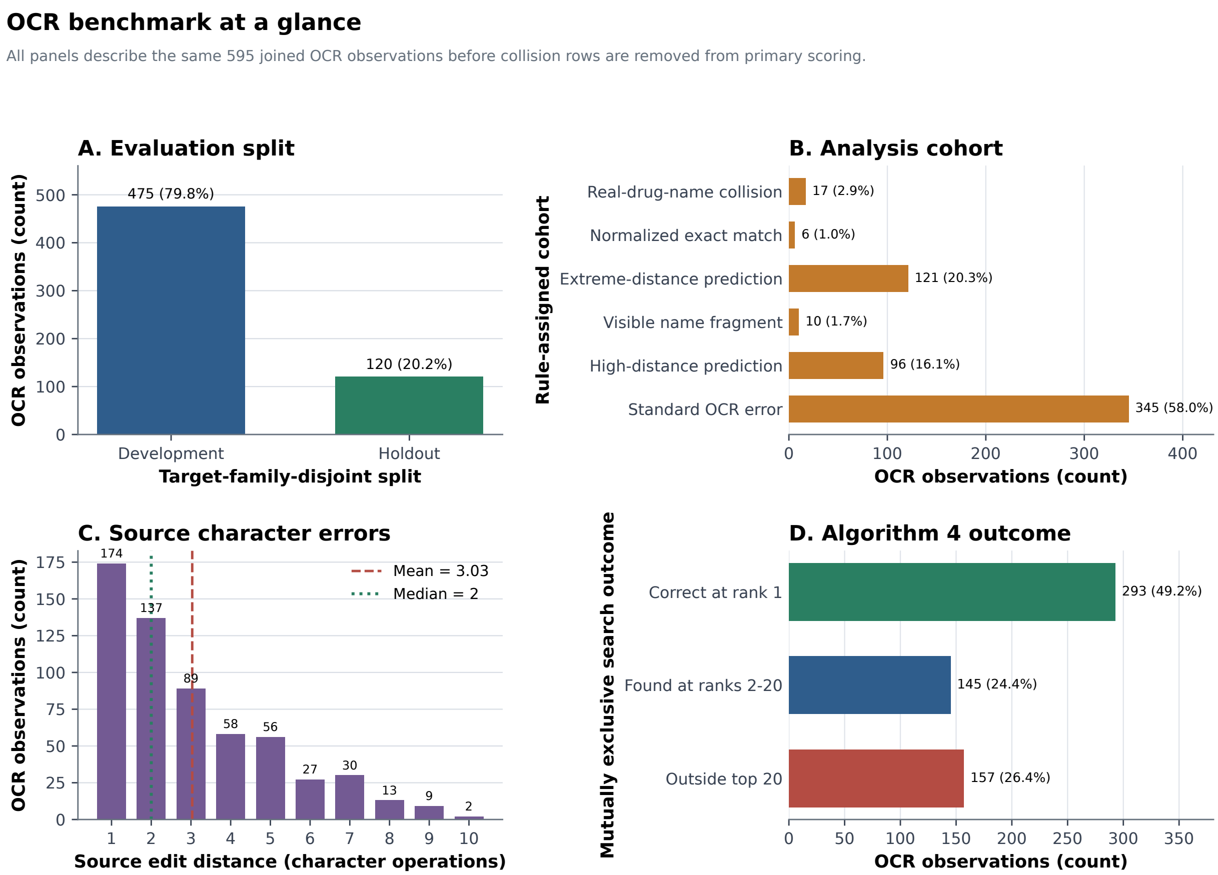

Figure 2: Four views of the same 595 OCR observations, explained panel by panel below.

- **Visible fragment** – 10 rows (1.7%): OCR only caught part of the word, but that part sits inside the target. Example: `RIVOTRI` vs `RIVOTRIL`.
- **High-distance prediction** – 96 rows (16.1%): moderately garbled, 40–60% of letters wrong. Example: `OSMOL` vs `OSTOCAL`.
- **Standard OCR error** – 345 rows (58.0%, the majority): ordinary, minor OCR mistakes, 40% or less wrong. Example: `ABISAGLOR` vs `ABASAGLAR`. This is the "typical" case the system should handle well.

Panel C – How garbled is the text. Most rows need only 1–2 character fixes (median 2); a handful of severe outliers (up to 10 fixes) pull the average up to 3.03.

Panel D – Outcome.

- **Correct at rank 1** – 293 rows (49.2%): first guess was right.
- **Found at ranks 2–20** – 145 rows (24.4%): right answer was on the list, just not first – a *ranking* problem.
- **Missed entirely (outside top 20)** – 157 rows (26.4%): right answer never showed up – a *retrieval* problem. Of these, 144 rows still returned wrong guesses and 13 returned nothing at all.

5 One row, start to finish

EasyOCR read a package as `MYMLAX`; the real medicine is `MYOLAX` – one letter is wrong. Algorithm 4 returned 36 candidates and ranked `MYOLAX` first. That is a full success. Below are six real rows showing every way this can go right or wrong:

- **`MYMLAX`** → **`MYOLAX`**, rank 1. *Full success* – one letter swapped, nailed on the first guess.
- **`KELOLAC`** → **`KETOROLAC`**, rank 2 (top guess was `KETOLAC`). *Ranking miss* – the system found it, just didn't put it first.
- **`RIV`** → **`RIVOTRIL`**, rank 10 (top guess `RIVO`). *Weak evidence* – only 3 visible letters, still enough to land in the top 20.
- **`GHONT`** → **`CALAMINE`**, missed entirely (top guess `GEOCAND`). *Retrieval miss* – OCR corruption wiped out nearly all real evidence.
- **`H`** → **`RIVO`**, missed, no candidates returned. *Insufficient evidence* – one character can't identify a medicine.
- **`RIVOTRIL`** → **`RIVO`**, missed (top guess `RIVOTRIL`, an exact different real drug). *Labeling conflict*, not a search failure.

6 How well does it work?

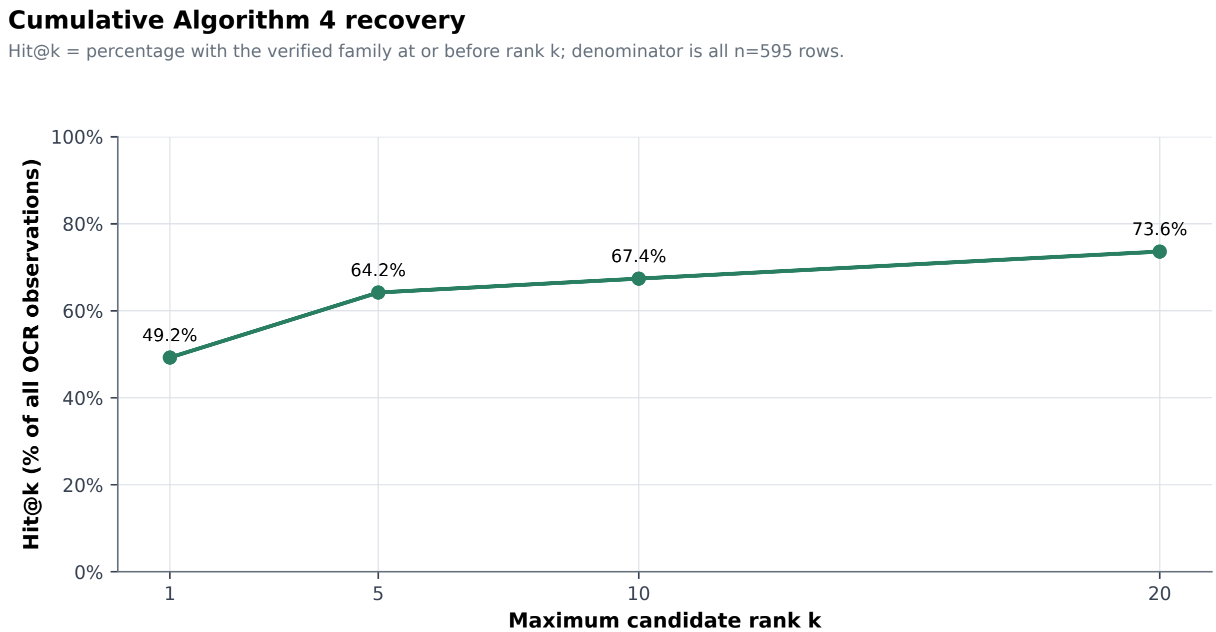

Figure 3: As you allow more guesses (rank cutoff), more correct answers show up. Recovery climbs from 49.2% at rank 1 to 73.6% at rank 20; rank 5 alone already captures 89 of the 145 rows that were missing at rank 1.

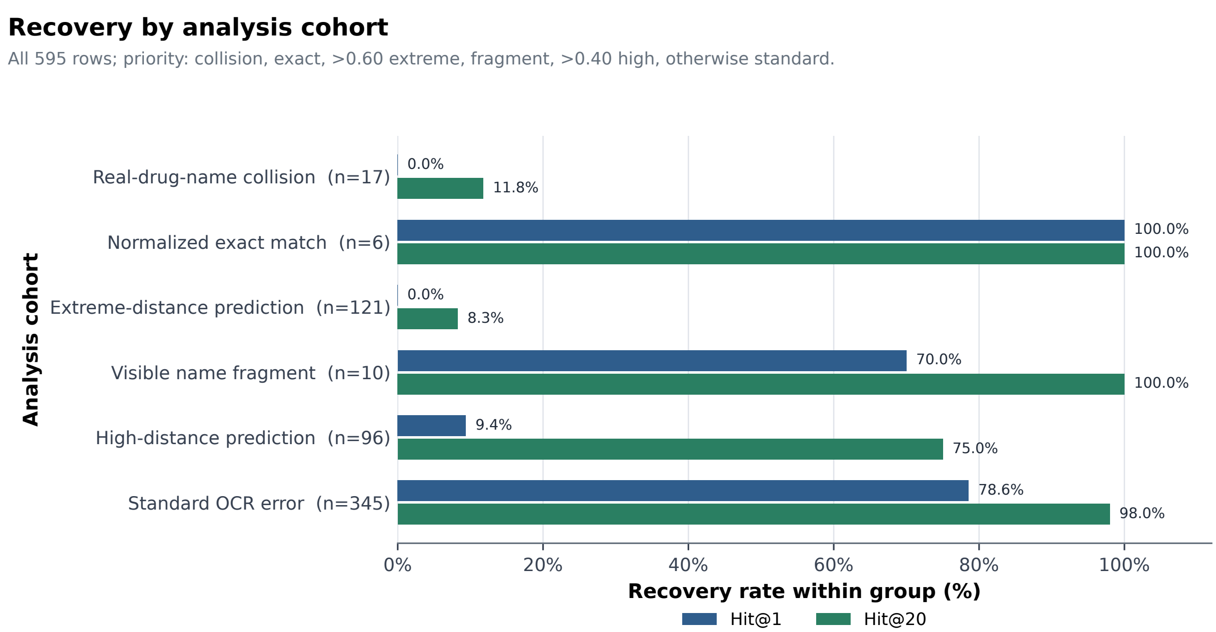

Figure 4: Recovery rate for each cohort from Section 4. Standard OCR errors reach 97.97% Hit@20 – the system is reliable on ordinary noise. Extreme-distance predictions reach only 8.26% Hit@20 – once the text is that corrupted, the search has almost nothing to work with.

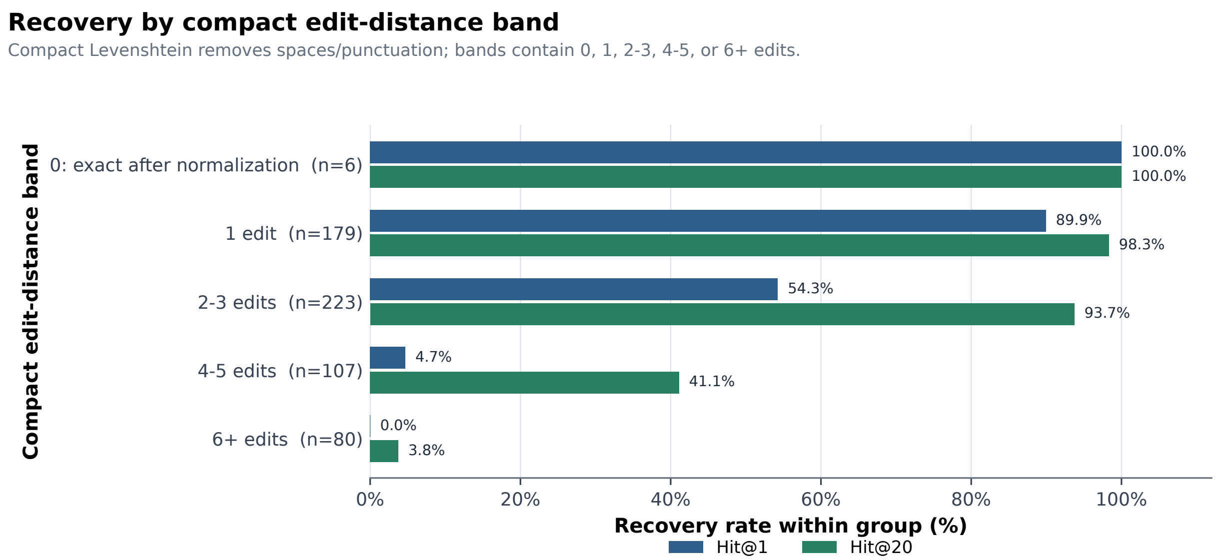

Figure 5: Recovery by number of character fixes needed. 1–3 fixes: still highly recoverable. 4–5 fixes: misses 63 of 107 rows (59%). 6+ fixes: misses 77 of 80 rows (96%) – essentially unrecoverable.

7 The hardest cases: badly garbled text

This is the 121-row group where more than 60% of the target's letters are wrong ($r > 0.60$) – 20.3% of the whole dataset.

- Average corruption in this group: $r = 0.937$ (median 0.833) – typically over 6 letters wrong out of a short medicine name.
- Outcome: 0 rows land at rank 1, 10 land somewhere in ranks 2–20, and 111 are missed entirely – 8.26% Hit@20.
- One OCR model, EasyOCR, supplies 46 of these 121 rows (38%) – a disproportionate share of the hardest cases.
- Four medicines – `MYOLAX`, `RIVOTRIL`, `RIVO`, `KETOROLAC` – account for 66 of the 121 rows (55%).

Rows worth double-checking before trusting the score:

- `ARDANCE` (labeled `RIVO`) → top guess `JARDIANCE`. This exact pair shows up across 7 different OCR models – looks like a possible mislabel.
- `CINNARIZIN` (labeled `MYOLAX`) → top guess `CINNARIZINE`, an almost-exact spelling of a different real drug.
- `INDOMETHACIN` (labeled `INDERAL`) → top guess `INDOMETHACIN`, an exact different real drug.
- `NEURONTIN` (labeled `NEUROCET`) → top guess `NEURONTIN`, same pattern.

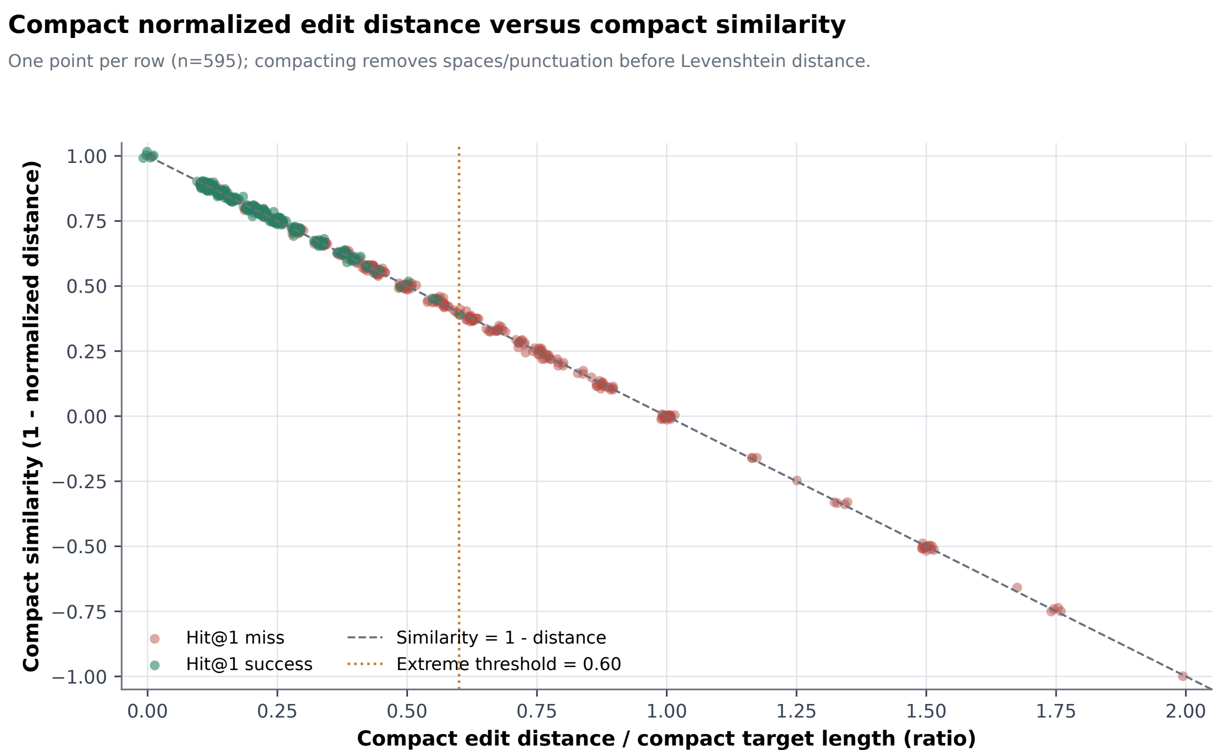

Figure 6: Every row plotted by how corrupted it is (right = more corrupted). Green = correct on first guess, pink = missed. Success clusters tightly on the low-corruption side of the 0.60 line; almost nothing succeeds past it.

8 What actually breaks recovery

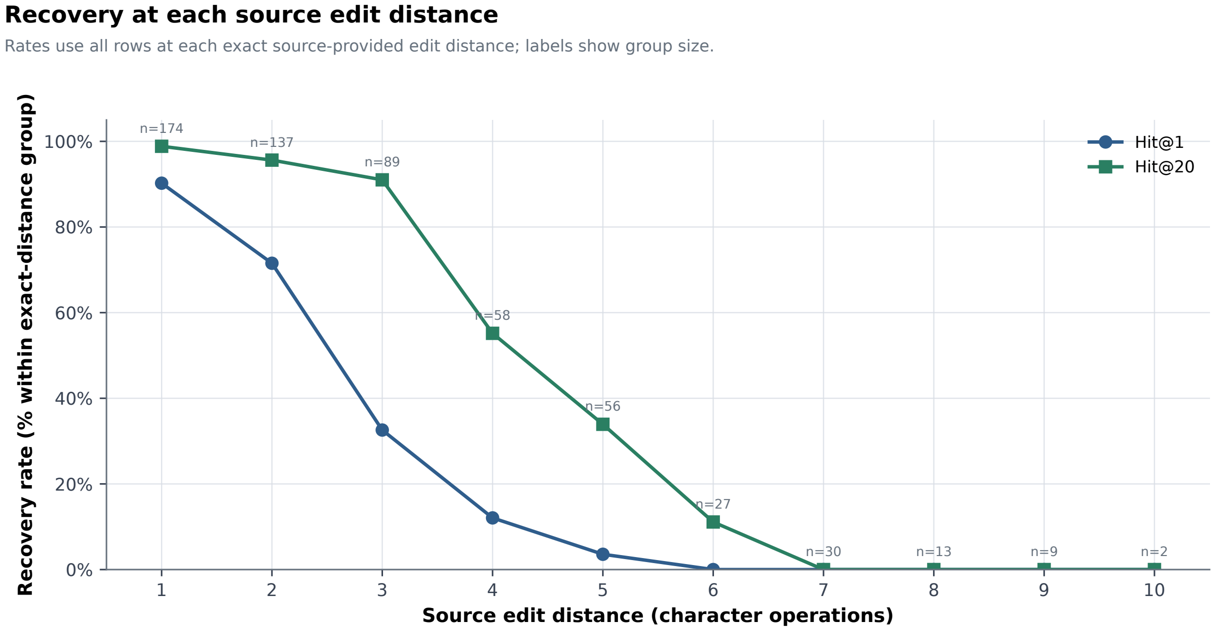

Figure 7: Recovery by exact number of character fixes. There's a cliff: Hit@20 is 91.01% at 3 fixes, drops to 55.17% at 4 fixes, and hits 0% by 7–10 fixes. **4 fixes is where recovery breaks down.** (Overall, more fixes correlates strongly with more failures: correlation −0.66.)

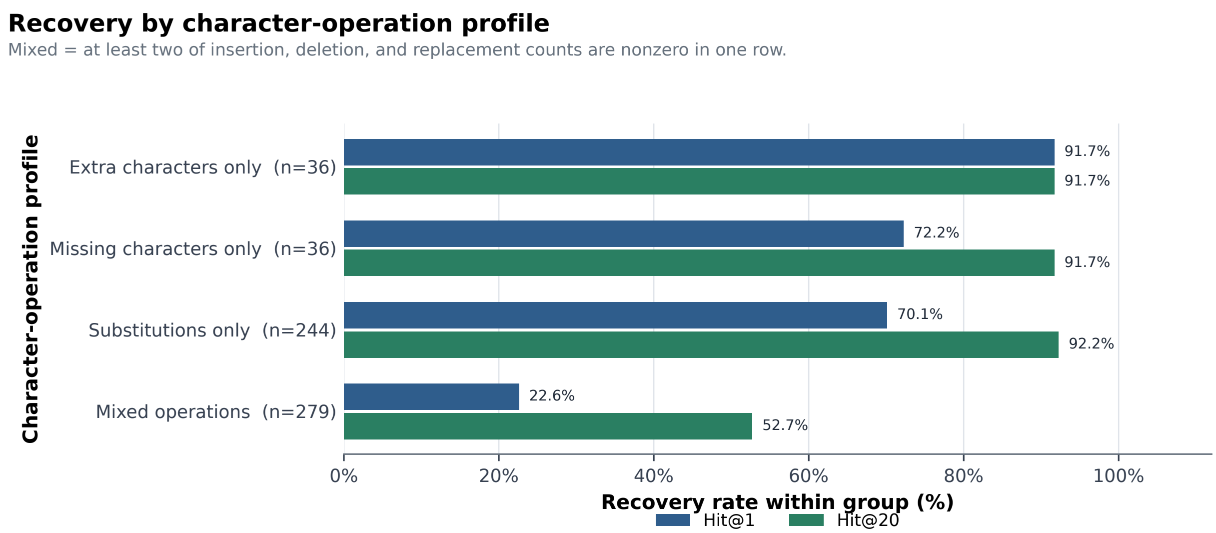

Figure 8: Rows with only *one* kind of mistake (only extra letters, only missing letters, or only wrong letters) reach 91.67%+ Hit@20 – easy. Rows with a *mix* of mistake types in the same word (279 rows, 47% of the dataset) reach only 52.69% Hit@20 – hard, and the single biggest fixable failure mode (see Section 10).

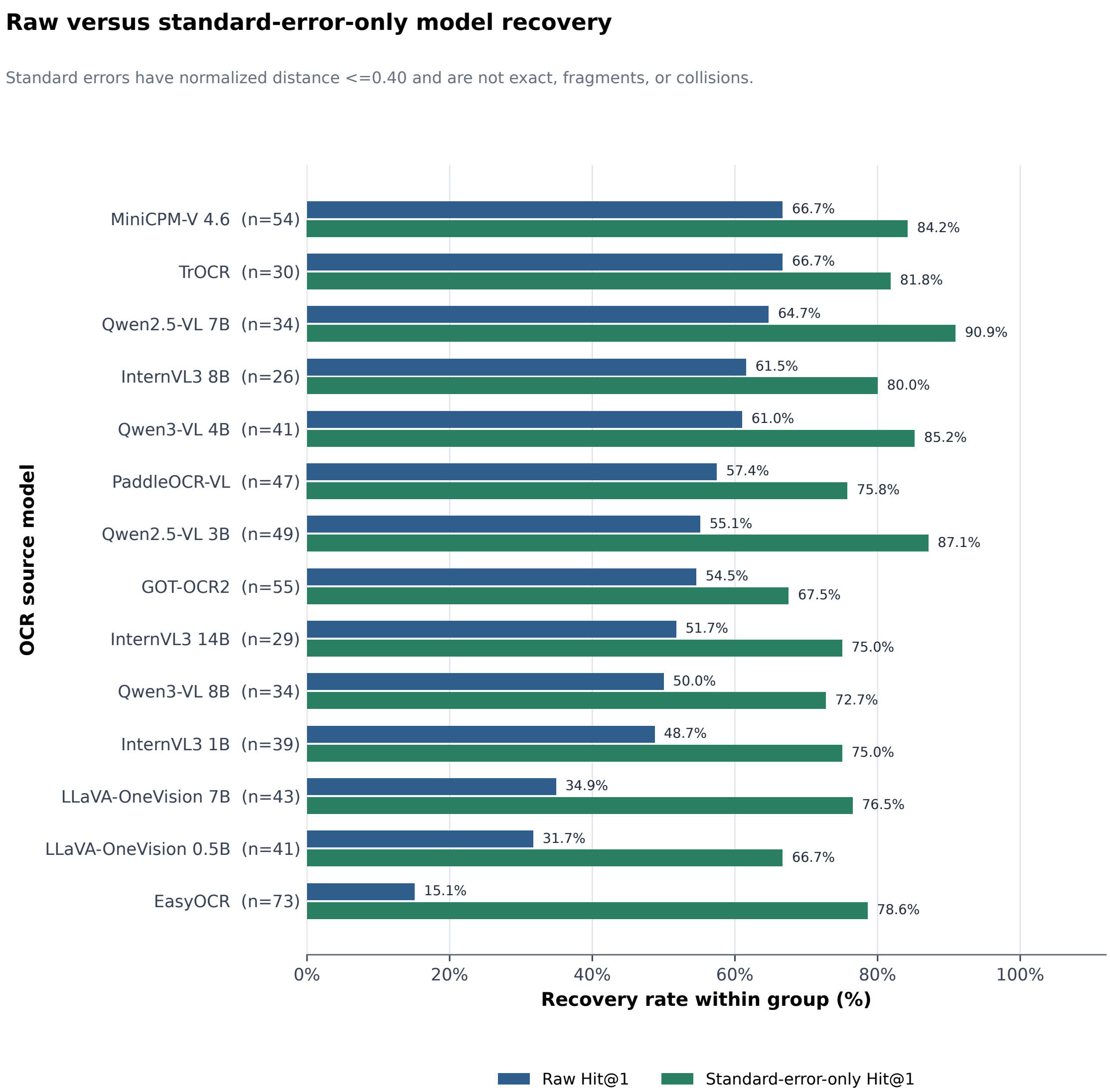

Figure 9: Raw model rankings are misleading. EasyOCR looks worst at 15.07% Hit@1 across all 73 of its rows – but 46 of those 73 (63%) happen to be extreme-distance cases. Once compared only on standard (mildly garbled) rows, EasyOCR jumps to 78.57% Hit@1, near the top. **Don't rank OCR models on this benchmark without controlling for how hard their images happened to be.**

9 Does it work on medicines it was never tuned on?

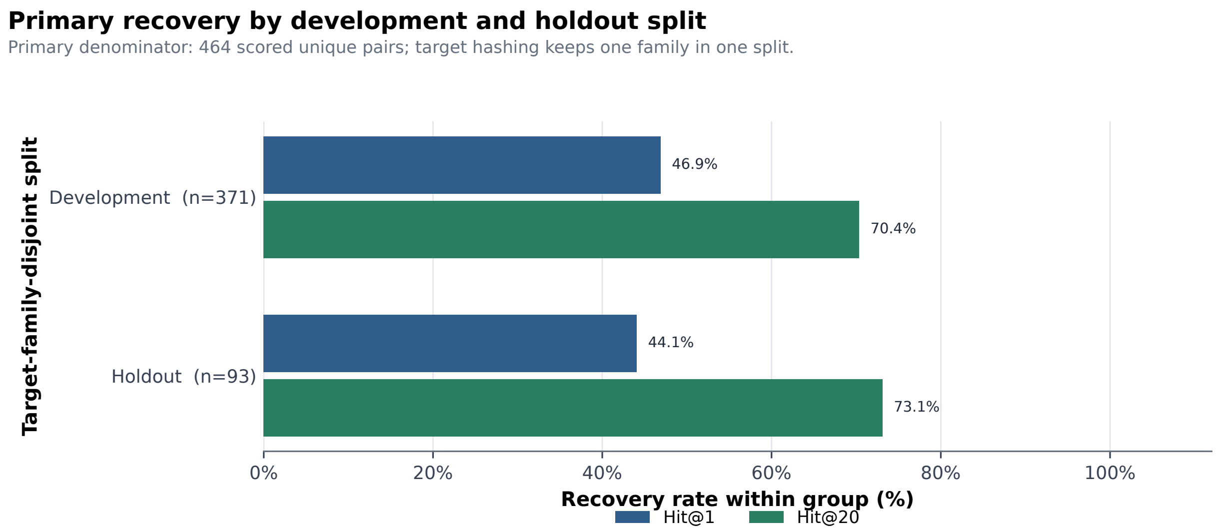

Figure 10: Development (371 unique pairs, includes `KETOROLAC`) vs. holdout (93 unique pairs, includes `VIGOREX` – a family excluded from all tuning).

- Holdout reaches 73.12% Hit@20 – close to the overall primary rate (70.91%). The system still *finds* medicine families it has never been tuned on almost as often as ones it has.
- Holdout Hit@1 is 44.09%, a bit below the overall 46.34%. **The accuracy gap on new medicines is mostly about ranking the right answer first, not failing to find it.**

10 Where the failures come from

Two different questions, two different bars in each figure below: *failure rate* (how often a group fails, among its own rows – tells you which group is hardest) vs. *failure share* (what fraction of all 157 misses that group causes – tells you where fixing effort pays off most).

Other failure sources, in brief:

- **Mixed-error rows** cause 132 of 157 misses (84%) – the largest single lever for improvement (matches Figure 8).
- **Very short OCR text** (1–3 characters): worst failure *rate* at 90% (18 of 20 miss), but the 6–7 character band causes more total misses (47) simply because it has far more rows (200).
- **By OCR model**: EasyOCR alone accounts for 32.48% of all misses (51 of its 73 rows), largely because it produced a disproportionate share of extreme-distance cases – not necessarily worse OCR quality.
- **By medicine**: `RIVOTRIL`, `MYOLAX`, `RIVO`, and `KETOROLAC` together cause 45.22% of all misses (71 of 157) – mostly because they have the most rows overall, not because they're uniquely hard.
- **Ranking-only misses** (145 cases where the right answer is at rank 2–20 but not rank 1): standard-error rows cause 67 of these and high-distance rows cause 63 – so a re-ranking fix would help both ordinary and moderately garbled cases, not just the extreme ones.

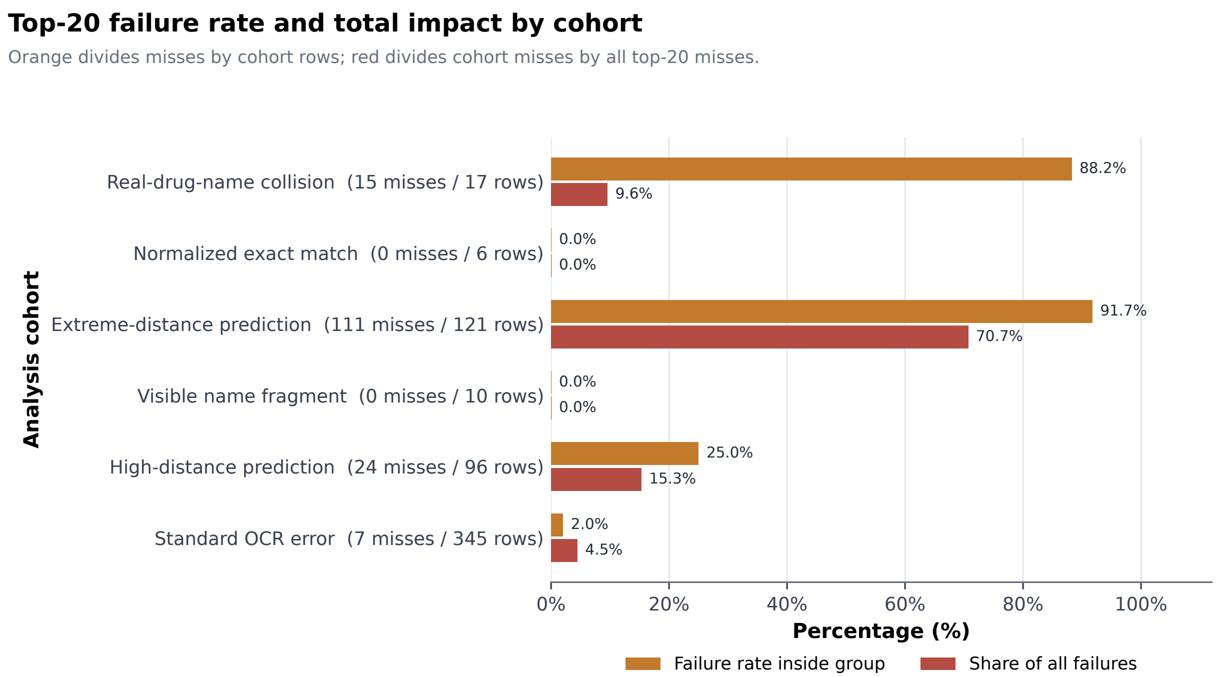

Figure 11: Extreme-distance rows are both the hardest *and* the biggest problem: 91.74% of them fail (111 of 121), and they alone account for 70.70% of every miss in the dataset.

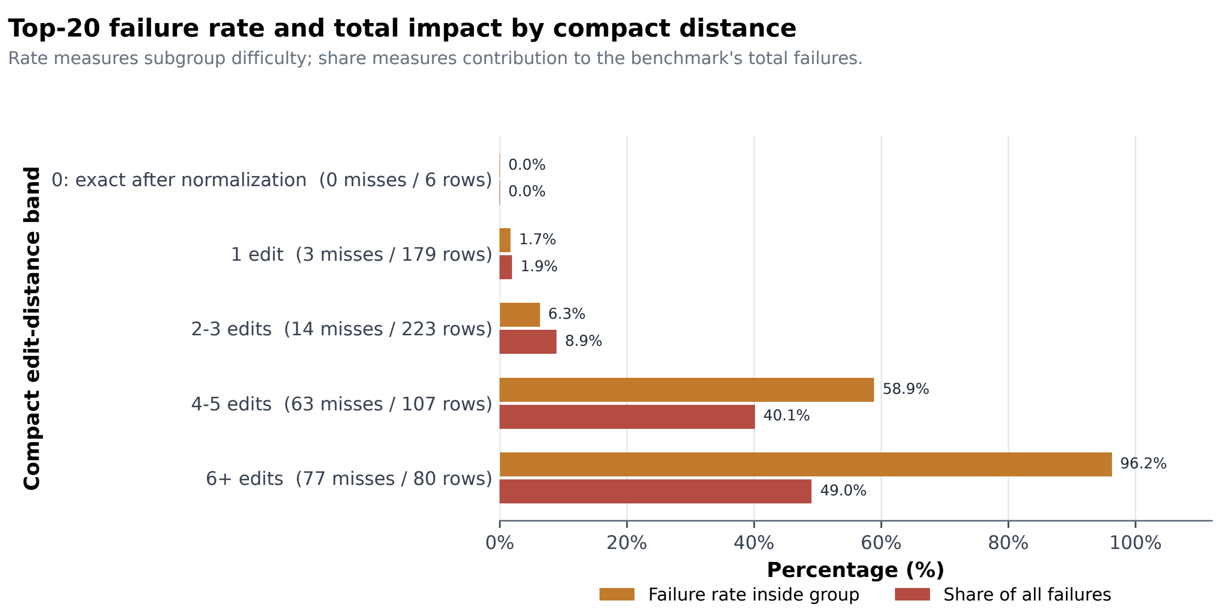

Figure 12: 4 or more character fixes cause 140 of the 157 misses (89.17% of all failures). The 6+ fixes band alone misses 77 of its 80 rows.

11 Is it safe?

- The system never confidently locks onto a wrong answer: 582 of 595 responses show a list of candidates rather than certifying one as correct, and the other 13 return nothing rather than guessing.
- Risk tags: SAFE 342 rows (57%), CAUTION 236 rows (40%), DANGEROUS 17 rows (3%, the real-drug-name collisions).
- **Zero rows** were confidently certified wrong – safety comes from showing candidates for review, not from blind top-1 answers.

12 What to fix, in order

P0 **Audit the collision/near-collision rows against the source images.** Several OCR outputs are exact spellings of a different real medicine, so the "correct answer" on file may itself be wrong. Do this before tuning anything else.

P1 **Improve ranking for positions 2–5.** 89 of the 145 rank-only misses would be fixed by pulling the right answer from rank 2–5 up to rank 1 – raises Hit@1 without touching retrieval or safety.

P2 **Add matching logic aware of mixed error types.** Rows with more than one kind of character mistake cause 84% of all misses – the biggest lever available.

P3 **Ask for more evidence on very short OCR text or suspected collisions** instead of forcing a guess.

P4 **Compare OCR models on the same set of images.** Raw model rankings are currently skewed by which models happened to get the hardest images (Figure 9).

13 Scope and source files

This benchmark tests medicine-*name* search after OCR only – it does not evaluate OCR accuracy itself, dosage extraction, clinical correctness, or whether every labeled answer is actually right (see Section 12, P0).

- Source observations: `benchmark_03_ocr/data/04_model_predictions/predictions.csv`
- Row-level search results: `benchmark_03_ocr/artifacts/04_model_predictions/algorithm_4_results.csv`
- Figure generator: `benchmark_03_ocr/build_model_prediction_paper_figures.py`
- All 40 generated figures (this summary uses 12 of them): `benchmark_03_ocr/results/04_model_predictions/figures/python_paper/`